

DATA 202 — Quiz 2

Week 4 — Covering Week 3: Cleaning, Grouping & Plots

Name: _____

Date: _____

The Dataset: Esports Players

`players` contains simulated records of competitive video game players, one row per player. The rows below are the **raw** data, exactly as loaded from the CSV. Question 1 works with this raw data; Questions 2–6 assume `players` has already been cleaned (standardized city names, ladder status, and skill levels).

Columns: `id`, `years_playing`, `sponsors`, `monthly_earnings_usd` (numerical) · `handle`, `city`, `ladder_status`, `skill_level` (categorical/text)

First few rows (raw — before any cleaning):

id	handle	city	ladder_status	years_playing	sponsors	monthly_earnings_usd	skill_level
15	kiRA	NEW-YORK	Ranked	14	4	579	ADVANCED
16	ZEPHYR	SEOUL.	ranked , pending	10	4	286	AMATEUR
35	echo	berlin	unranked	9	4	375	EXPERT
45	mochi.	los Angeles	RANK	5	2	90	novice
49	NEO	toronto	CASUAL	10	1	399	ADVANCED

Question 1 (SLO 03A)

The code below is run on the `ladder_status` column of the **raw** data shown above.

```
players["ladder_status"] = (  
    players["ladder_status"]  
    .str.strip()  
    .str.lower()  
    .str.replace(r"ranked\s*,\s*pending", "ranked pending", regex=True)  
)
```

What will this code do to the `ladder_status` column? Explain what each step does, and say what the `ladder_status` values in the rows above will look like afterward.

Question 2 (SLO 03B)

Complete the code below to compute, for each `skill_level`, both the number of players and the average `years_playing`, using named aggregation.

Then answer: if you used `"median"` instead of `"mean"` for `avg_years`, would the result be more or less sensitive to one extremely high outlier in `years_playing`?

```
result = players.groupby("_____").agg(  
    n_players=("id", "count"),  
    avg_years=("_____", "_____"),  
)
```

Question 3 (SLO 03C)

For each question below about the `players` data, name the best plot type (Histogram / Scatter / Line / Bar) and which column(s) you'd map to x (and y, if relevant).

Question	Plot Type	Column(s)
What's the distribution of <code>years_playing</code> across all players?		
How do <code>sponsors</code> and <code>monthly_earnings_usd</code> relate to each other?		
How does average <code>sponsors</code> change as <code>years_playing</code> increases?		
How does average <code>years_playing</code> compare across <code>ladder_status</code> values?		

Question 4 (SLO 02A) — Review from Quiz 1

For each command below, identify whether it **Accesses**, **Adds**, or **Deletes** data, and whether it operates on a **Row** or a **Column**.

#	Command	Access / Add / Delete?	Row(s) or Column?
1	<code>players["years_playing"]</code>		
2	<code>players["veteran"] = players["years_playing"] > 10</code>		
3	<code>players = pd.concat([players, pd.DataFrame([new_row])], ignore_index=True)</code>		
4	<code>players = players[players["city"] != "Berlin"]</code>		
5	<code>players = players.drop(columns=["sponsors"])</code>		

Question 5 (SLO 02B) — Review from Quiz 1

Complete the code below to select all players in “**Seoul**” with `monthly_earnings_usd` below 100, sorted by `monthly_earnings_usd` from lowest to highest.

Remember: the filter condition still goes inside the square brackets and must reference `players` again, e.g. `players[players["Column"] ...]`.

```
result = players[
    _____
]
result = result.sort_values("_____", ascending=_____)
print(len(result))
```

Question 6 (SLO 02C) — Review from Quiz 1

Available visual channels: `x`, `y`, `color`, `size`, `symbol`, `facet_col`.

As on Quiz 1: since this is a paper quiz, `color` isn't something you can sketch in pencil — the code below uses `symbol` instead, mapping a categorical column to different *marker shapes* (e.g. $\times$, $\circ$, $\triangle$) rather than colors.

For the code below, fill in the table: for each visual channel it uses, name the column mapped to it and say whether that column is Numerical or Categorical.

```
px.scatter(players, x="sponsors", y="monthly_earnings_usd",
           symbol="skill_level", size="years_playing",
           title="Sponsors vs. Monthly Earnings")
```

Visual Channel	Column Used	Data Type (Numerical / Categorical)
x-axis		
y-axis		
symbol		
size		

In the space below, sketch what you expect this chart to look like. Label your axes. You don't need exact data — just show the general pattern (point cloud, marker shapes ($\times/\circ/\triangle$) for groups, varying point sizes).